
Register, Not Procedure: What a Conditional Linear Map Can and Cannot Install

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Abstract

Activation steering and parameter-efficient finetuning both edit a model’s behavior, and both can be written as an *input-conditional affine map* of the residual stream, $h \mapsto h + (Wh + b)$. We ask where the *same* map family succeeds and where it fails, holding the apparatus fixed across tasks. The boundary is sharp and reproducible: the map installs and transports a **register/disposition** (refusal *tone*, a yes/no answer disposition) but never the trajectory scaffold of a multi-step **procedure**—on either procedure we test, GSM8K arithmetic and open-book multi-hop QA. A primal-ridge conditional map dominates every fixed-vector steering baseline on the harm-refusal frontier, yet the identical machinery recovers ≈ 0 of a reasoning LoRA’s gain—and so do MLP, on-policy DAGger, per-context, and task-loss (DAS) variants. We trace the failure to its mechanism: the reasoning capability is a *distributed, time-dense trajectory state* (recovery needs $\geq 94\%$ of decode steps and $\sim 80\%$ of the residual’s variance band), not a low-rank point-wise function of the activation. That mechanism is not arithmetic-specific: on the multi-hop procedure the same layer-20 lockstep oracle recovers the same fraction of the budget (+0.76 vs 0.75) with the same temporal-density signature, and a teacher-forced gold-token lens reproduces the *ordering* of the deficit (planning tokens are the base’s worst class) while showing that what the base retains per step is task-specific: multi-hop “execution” is retrieval from context, not computation, and nothing in its chain crystallizes late in the stack the way an arithmetic result does. Cross-model, even a *trained* register edit carries only the coarse disposition off-policy; a strictly better-fitting affine bridge transports it *worse*, because the edit lives off-manifold and regression shrinks exactly what must be preserved. We propose the register/procedure distinction as the organizing axis for “what is steerable,” and outline the experiments that would turn it from an empirical regularity into a predictive criterion. All results are on the Llama-2-7B family.

1 Setup and question

A behavior can be controlled by adding a vector to activations (CAA, Arditi, ActAdd, CAST) or by finetuning a low-rank edit into the weights (LoRA) or a learned subspace (LoReFT). LoReFT’s edit $h + (Wh + b - hR)R^\top$ and our primal-ridge steering map $(I + \alpha W)a$ —fit closed-form so that $Wa \approx \delta = a_{\text{expert}} - a_{\text{base}}$ —are the *same object*: an input-conditional affine edit in a learned subspace. We fix this object and sweep the **task**, asking a single question—*does the capability install?*—under one coherence-aware metric. Two behaviors anchor the extremes: refusal (a disposition) and GSM8K chain-of-thought arithmetic (a procedure). Commonsense QA sits between them. Because a single procedure cannot distinguish “procedures do not install” from “arithmetic does not install,” we then re-run the entire procedure apparatus on a second one: open-book multi-hop QA over MuSiQue, where a matched LoRA takes the base from 0.000 to 0.63 and the model must *compose* facts laid out in the prompt rather than compute over numbers in it.

2 Findings

(R) Register installs from the conditional map alone, dominating fixed vectors. Pointing the ridge map at refusal and scoring on the harm-refusal / over-refusal Pareto frontier (60+60 held-out prompts, refusal tone conditioned on coherence), only the *input-conditional* map reaches the harmful-refusal axis at all (Table 1). The fixed vectors go straight from no-effect to off-manifold gibberish (their per-layer ℓ_2 norm is ~ 49); the map’s per-input shift lands *on* the chat manifold and even repairs base coherence ($0.63 \rightarrow 0.98$) before it refuses. Conditionality is learned, not bolted on: Wa is large on harmful inputs and ≈ 0 on benign. The δ it regresses is near-perfectly decodable (held-out $R^2 \approx 0.9997$), so the open question was never “can δ be read off” but “does injecting it steer”—and for register it does.

Table 1: Best harmful-refusal at benign over-refusal ≤ 0.10 . Only the conditional map reaches the refusal axis.

method	conditionality	harmful refusal	benign over-refusal
CAA mean-difference vector	none	0.00	—
Arditi single direction	none	0.00	—
CAA + logistic gate (CAST)	explicit classifier	0.00	—
ridge map Wa ($\alpha=0.12$)	learned, for free	0.62	0.02

(R) Commonsense is a register both LoRA and LoReFT solve—via divergent routes. A paper-faithful LoReFT (32 layers, rank 8, f7+l7) and a matched LoRA trained on the *byte-identical* supervised signal both recover the task from a 0.00 base (BoolQ/PIQA/ARC-C: LoReFT 0.667/0.798/0.602, LoRA 0.680/0.790/0.655). But the *edits* are not the same change: CKA(δ) collapses 0.96 (L4) $\rightarrow 0.13$ (L28), per-token cosine stays 0.11–0.39, and top-8 subspace overlap stays 0.16–0.25. They share only an early-layer, rotation-equivalent skeleton (the register/format nudge); the task content is written through genuinely different deep-layer edits. There is no canonical “commonsense direction.”

(P) Procedure does not install—under any map we tried. The same machinery pointed at GSM8K recovers ≈ 0 of a MetaMath-LoRA’s budget, robustly across the obvious salvages (Table 2). The map installs reasoning-*shaped* tokens with no arithmetic underneath—“the arithmetic wall.” On the second procedure the wall is present but **lower**: the nonlinear and on-policy rungs still collapse to 0, yet the *linear* rung recovers $\sim \frac{1}{4}$ of the multi-hop budget at L20 and $\sim \frac{1}{2}$ at L24 (a narrow $\alpha = 1.0$ resonance). Whether that is a genuine task difference is being measured directly: GSM8K’s ridge map was never probed per-layer at L20 or L24, so the two curves are not yet matched (Section 4).

Table 2: Procedure-null lineup: every salvage of the conditional map recovers ≈ 0 on GSM8K.

salvage attempt	result	rules out
global primal-ridge map	0.00	— (baseline null)
short-output single-operation arithmetic	0.00	trajectory <i>length</i> as the wall
per-context local refit (problem-specific map)	0.00	map <i>generality</i>
on-policy DAgger refit ($\approx 97\%$ on-policy by round 2)	0.00	<i>distribution shift</i>
DAS task-loss subspace (CE $\rightarrow$ 0.038, $r=512$)	0.00	low-rank-low-variance <i>subspace search</i>

(P) The procedure is a distributed, time-dense trajectory state. A *lockstep oracle* that overwrites the base’s L20 residual with the LoRA’s every step *does* recover GSM8K (0.75), which let us dissect what the map cannot supply. (i) **Distributed, not transported:** identity-ablating just the top two downstream layers (30–31) craters not only the patched base ($0.75 \rightarrow 0.10$) but the natively-capable LoRA itself ($1.00 \rightarrow 0.10$)—L20 holds an *intermediate*, not a finished answer. (ii) **Compute is not the deficit:** teacher-forced on the correct chain, the base predicts computed-result tokens at 96.8%; its failure is *trajectory control*, not per-step arithmetic. (iii) **Dense in time and in variance:** recovery requires patching $\geq 94\%$ of decode steps (every $\leq 50\%$ gate $\rightarrow 0.00$) and $\sim 80\%$ of δ ’s variance band (top-64 = 55% of energy but 0% recovery; the cliff is top-256 $\rightarrow$ 512). A donor-free steerer would therefore have to reproduce the LoRA’s near-exact 4096-d state at almost every step—i.e., *be the donor*. The procedure is not a low-rank, pointwise, or time-sparse function of the activation.

(G) The procedure findings are structural, not arithmetic-specific. One procedure cannot separate “procedures do not install” from “arithmetic does not install,” so we re-ran the apparatus end-to-end on MuSiQue open-book multi-hop QA (matched LoRA 0.000 $\rightarrow$ 0.63; 317 base-fail/donor-solve contrast problems). The two structural axes replicate exactly. *Oracle*: the lockstep oracle at the same layer recovers +0.76 (GSM8K 0.75), all-layer control +1.00. *Temporal density*: patching every other step collapses recovery to 0.05, every sparser periodic gate is ≈ 0 , and the structural complement—patch everything except the answer span—equals the full oracle while firing at $\sim 100\%$ of steps, the same signature GSM8K gives. *Plan vs execute*: teacher-forcing the base on the gold chain and lensing the gold token by its role, the base agrees with **execution** tokens more than **planning** tokens (+0.055 [+0.040, +0.069], bootstrapped over problems), so the trajectory-control reading survives the task change—and supplying the trajectory makes the *later*, nominally-composed hops **easier** than the first (+0.130), which is what a trajectory deficit predicts and a per-step composition deficit does not. What does *not* carry over is the shape: where GSM8K’s computed results sit +0.133 [+0.096, +0.173] above the chain average and crystallize with depth (lens rank 18 $\rightarrow$ 7 $\rightarrow$ 0 over L20–24), multi-hop execution merely *matches* the chain average (+0.001, interval spanning 0) and no role crystallizes at all—its hop answers are copies from the in-context passages (the final restatement scores 0.933 at lens rank 0 in every layer). **The trajectory-control deficit is general; the per-step work the base retains is task-specific, and only on arithmetic is it computation rather than retrieval.**

(X) Cross-model, even a trained register edit carries only the coarse disposition—and aiming better hurts. Transplanting a trained LoReFT edit from a donor (Llama-2-7B base) to a recipient (Llama-2-7B-chat) through a per-layer orthogonal-Procrustes bridge transfers the *disposition* off-policy (BoolQ boolean-commitment 32 $\rightarrow$ 53/64, lenient 0.33 $\rightarrow$ 0.58) but not the trained output template or answer correctness (exact-match ≈ 0). On-policy refit is *worse*; and a ridge-fit **affine** bridge with 12–45 $\times$ better held-out geometry transports the edit *worse still* (8/64), because the LoReFT delta lives off the data manifold ($\|\delta\| > \|h\|$) and regression shrinks weakly-represented directions—precisely the ones that must be preserved. **Strength beats aim**: full-norm deltas through a crude isometry move behavior; well-aimed deltas at half norm do nothing. The geometric “wall” was real but behaviorally irrelevant.

3 Synthesis

One axis organizes all of the above. A **register/disposition**—refusal tone, an answer format, a yes/no commitment—is a low-rank, on-manifold, roughly *pointwise* function of the activation; it is installable by a single conditional affine map and even survives a crude cross-model rotation. A **procedure**—multi-step arithmetic—is a high-rank, off-manifold, *time-dense* trajectory state; it is invisible to variance- and task-loss-based subspace search, and recoverable only by transporting essentially the donor’s full per-step state. Finetuning (LoRA/LoReFT) and activation steering are the same map family, and they inherit the same boundary: they relocate *where a disposition points*, not *how a computation unfolds*. The second procedure sharpens which half of that is general. What transfers unchanged is the *trajectory* claim: the recoverable state sits at the same layer, is dense in the same way, and the base’s deficit is again control of the chain rather than the work inside a step. What does not transfer is the *content* of a step—arithmetic makes the base compute a result late in the stack, multi-hop only makes it copy one from context—and, correspondingly, how much of the edit a linear map can carry. So “procedure” is not one thing: it is a trajectory scaffold, which the map family never installs, wrapped around per-step work whose nature (and installability) is task-dependent. A corollary for distillation: a trained edit can be “distilled” by the conditional map only when fit on the steered (on-policy) input—but that is near-tautological (the edit is affine in its own input), a positive control, not an independent capability-install.

4 Next steps

- **Match the ladder axis across tasks (in progress).** The one place the two procedures appear to differ is the linear rung, and that comparison is not yet matched: GSM8K’s ridge map was probed per-layer only at L0/L1/L14/L16/L31 (all 0.00) and under all-layer joint injection, never at L20 or L24—the two layers where the multi-hop leak appears. We are refitting the GSM8K maps and running the identical per-layer protocol. If GSM8K reads ≈ 0 there, the task-dependent-

core claim stands; if it also leaks, the divergence collapses and the linear rung replicates like the others.

- **A third procedure** to move beyond $n=2$, ideally one whose per-step work is neither arithmetic nor retrieval, to test whether “trajectory general, per-step task-specific” holds as stated.
- **Hybrid transplant bridge**—affine forward map (better-aimed donor delta) + *orthogonal* back-projection (norm-preserving transport), to isolate aim from strength; our transport analysis predicts it beats the rigid-rotation baseline.
- **Quantify the disposition carry** beyond $n=64$ smoke: loosen the extractor to bare true/false, re-score full BoolQ/PIQA/ARC splits.
- **Same-family base→base transplant** to separate “cross-model” from “base→chat instruction-tuning shift” as the cause of the carry failure.
- **A register-vs-procedure predictor**: turn the empirical boundary into a measurable criterion—e.g. does (δ -rank, off-manifold fraction $\|\delta\|/\|h\|$, temporal density of the recovery gate) predict installability *a priori* on a held-out behavior (tone, sentiment, sycophancy, induction)?
- **Leak diagnostic for installed refusal**: the tone axis is established; run the two-axis HarmBench leak test on a compliant instruction-following base (e.g. Vicuna) to test whether content is withheld under the installed apology.

Evidence index. Refusal frontier, LoRA-vs-LoReFT similarity, the procedure-null lineup (short-output / local-refit / DAgger / PCA-band / DAS), the L20 oracle dissection (downstream lesion, gold-token lens, temporal gate), the cross-model transplant, and the multi-hop generality study (oracle, ladder, temporal density, plan-vs-execute) are each documented as standalone reports under [results/attribution/](#) with seeded, CPU-tested, reproducible drivers.